



Branding is what people say about you when you are not in the room.



-Jeff Bezos



DO-IT-YOURSELF

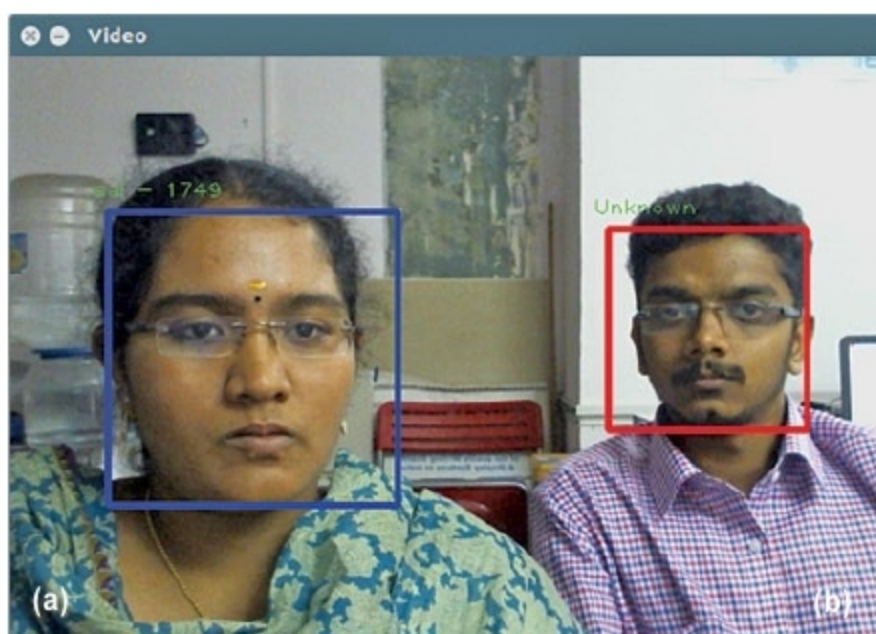


Fig. 2: (a) Trained face recognised along with confidence value and (b) Untrained face recognised as unknown

Download all relevant files, including main face recognition program (gui.py), written in Python, from the source code folder available on source.efymag.com. Then, extract the folder to your system, where you will find an empty folder called datasets. Browse the extracted folder from the terminal to run the face recognition program as explained in the testing steps below.

Testing steps

The steps are as follows:

1. Create a subfolder (say, sai) inside the datasets folder for the user whose face is to be trained.
2. Run Python-based face recognition program. For that, type the command given below.

```
$ python gui.py
```

You will see a welcome GUI window as shown in Fig. 1.

In the GUI, type the same name (sai) created as a subfolder inside the datasets folder. Press 'train' to start the face training program. The camera app will turn on and the program will start to capture video frames of the face.

You can sit in front of the camera while it is capturing the frames during training. It captures 100 photos with different facial expressions. The reason for capturing 100 frames is

to make sure that values of different facial expressions shown by the user are properly stored in the program, and to make the program easily recognise and retrieve the user's face in future. But note that, only the front of the face (facing the camera) is considered in this algorithm.

3. Once about 100 frames of the face have been captured, eigen_train_face.xml file will be generated automatically, where you will find the values of the faces being captured. Then, to close train program, press q on the keyboard while the camera app is still running.

(When you want to train again, the xml file will not be recreated, it will just get updated.)

4. Once the camera app is closed, press 'recognize' and the camera will start to recognise the faces from the database of captured frames. That is, the program will start matching the current face with the faces stored in the database. If a match is found, it will match it with the user's name along with the confidence value (Fig. 2).

If a match is not found, it will automatically display 'unknown' on the screen. Numerals depicted along with the name show the confidence level of the training, which, in turn, depends on various parameters like lighting, clarity of frames taken, and usage of right haar cascade face detection config and .xml files.

Training should be done for one person at a time so that the program can train and recognise the faces efficiently. This program can be further improved with more extraction features and added facilities. **EFY**

EFY Note

The source code of this project is available for free download at source.efymag.com



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